9.Writing GUIs in Python Using TKinter

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Getting Started with Tkinter

1. Import tkinter package and all of its modules.
2. Create a root window. Give the root window a title(using **title()**) and dimension(using **geometry()**). All other widgets will be inside the root window.
3. Use **mainloop()** to call the endless loop of the window. If you forget to call this nothing will appear to the user. The window will wait for any user interaction till we close it.

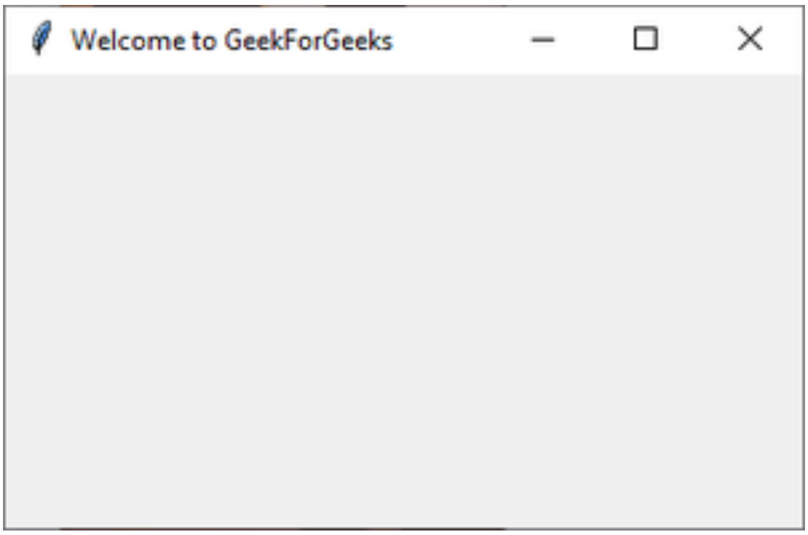
```
# Import Module
from tkinter import *

# create root window
root = Tk()

# root window title and dimension
root.title("Welcome to GeekForGeeks")
# Set geometry (widthxheight)
root.geometry('350x200')

# all widgets will be here
# Execute Tkinter
root.mainloop()
```

Output



Root Window

4. We'll add a label using the Label Class and change its text configuration as desired. The **grid()** function is a geometry manager which keeps the label in the desired location inside the window. If no parameters are mentioned by default it will place it in the empty cell; that is 0,0 as that is the first location.

```
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from tkinter import *

# create root window
root = Tk()

# root window title and dimension
root.title("Welcome to GeekForGeeks")
# Set geometry(widthxheight)
root.geometry('350x200')

#adding a label to the root window
lbl = Label(root, text = "Are you a Geek?")
lbl.grid()

# Execute Tkinter
root.mainloop()
```
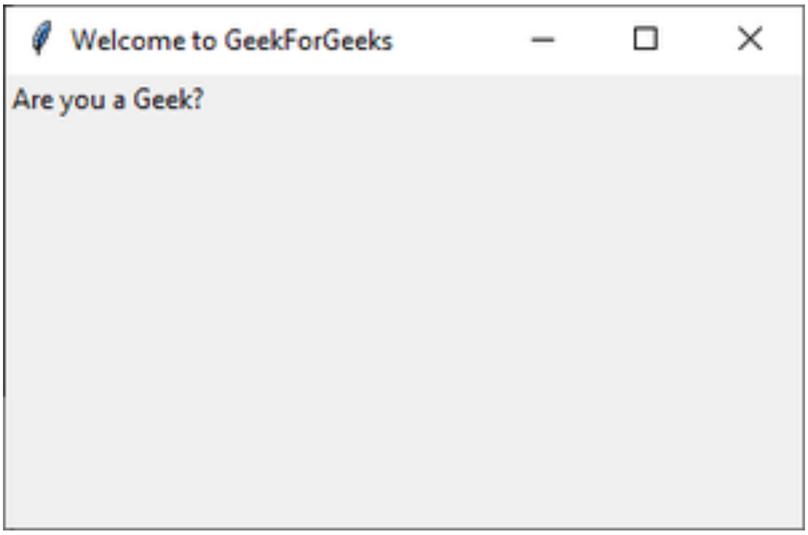

Output



Label inside root window

5. Now add a button to the root window. Changing the button configurations gives us a lot of options. In this example we will make the button display a text once it is clicked and also change the color of the text inside the button.

```python
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from tkinter import *

# create root window
root = Tk()

# root window title and dimension
root.title("Welcome to GeekForGeeks")
# Set geometry(widthxheight)
root.geometry('350x200')

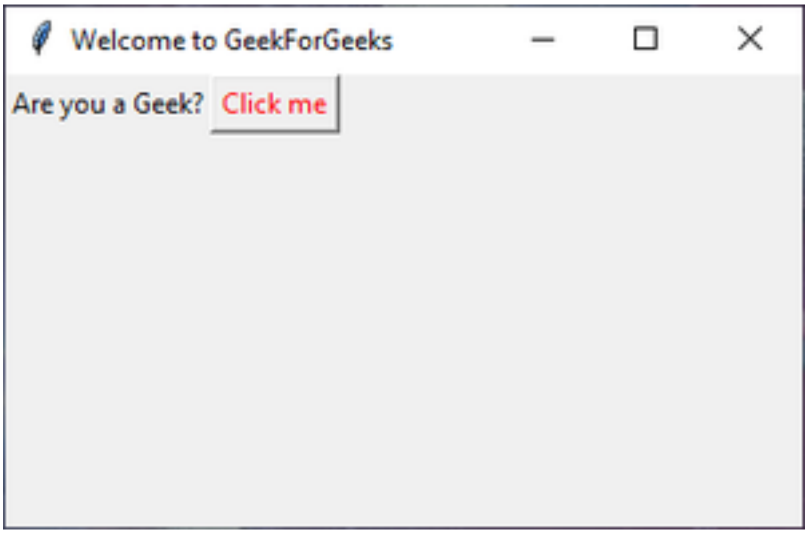
# adding a label to the root window
lbl = Label(root, text = "Are you a Geek?")
lbl.grid()

# function to display text when
# button is clicked
def clicked():
    lbl.configure(text = "I just got clicked")

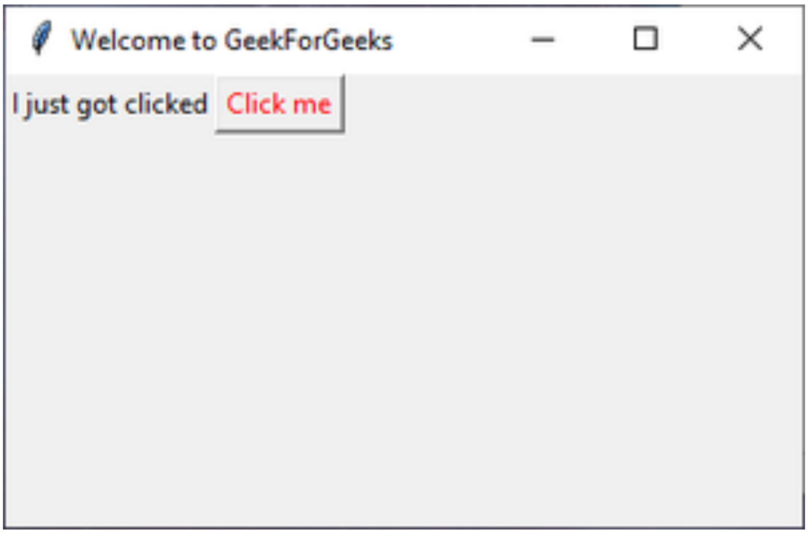
# button widget with red color text
# inside
btn = Button(root, text = "Click me" ,
             fg = "red", command=clicked)
# set Button grid
btn.grid(column=1, row=0)

# Execute Tkinter
root.mainloop()
```

Output:



Button added



After clicking "Click me"

6. Using the **Entry()** class we will create a text box for user input. To display the user input text, we'll make changes to the function **clicked()**. We can get the user entered text using the **get()** function. When the **Button** after entering of the text, a default text concatenated with the user text. Also change button grid location to column 2 as **Entry()** will be column 1.

```
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from tkinter import *

# create root window
root = Tk()

# root window title and dimension
root.title("Welcome to GeekForGeeks")
# Set geometry(widthxheight)
root.geometry('350x200')

# adding a label to the root window
lbl = Label(root, text = "Are you a Geek?")
lbl.grid()

# adding Entry Field
txt = Entry(root, width=10)
txt.grid(column =1, row =0)
```

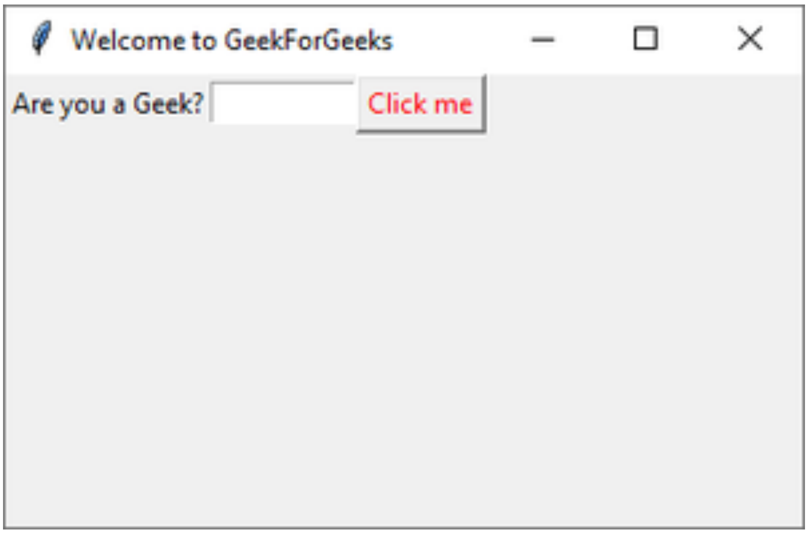
```
# function to display user text when
# button is clicked
def clicked():

    res = "You wrote" + txt.get()
    lbl.configure(text = res)

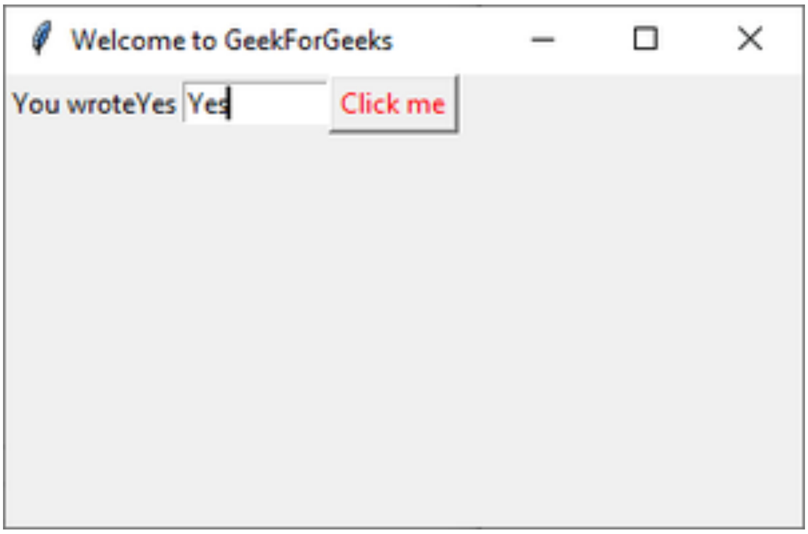
# button widget with red color text inside
btn = Button(root, text = "Click me" ,
             fg = "red", command=clicked)
# Set Button Grid
btn.grid(column=2, row=0)

# Execute Tkinter
root.mainloop()
```

Output:



Entry Widget at column 2 row 0



Displaying user input text.

7. To add a menu bar, you can use **Menu** class. First, we create a menu, then we add our first label, and finally, we assign the menu to our window. We can add menu items under any menu by using **add_cascade().**

```
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# create root window
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# root window title and dimension
root.title("Welcome to GeekForGeeks")
# Set geometry(widthxheight)
root.geometry('350x200')

# adding menu bar in root window
# new item in menu bar labelled as 'New'
# adding more items in the menu bar
menu = Menu(root)
item = Menu(menu)
item.add_command(label='New')
menu.add_cascade(label='File', menu=item)
root.config(menu=menu)

# adding a label to the root window
lbl = Label(root, text = "Are you a Geek?")
lbl.grid()

# adding Entry Field
txt = Entry(root, width=10)
txt.grid(column =1, row =0)


# function to display user text when
# button is clicked
def clicked():

    res = "You wrote" + txt.get()
    lbl.configure(text = res)

# button widget with red color text inside
btn = Button(root, text = "Click me" ,
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btn.grid(column=2, row=0)
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```
# Execute Tkinter
root.mainloop()
```